# AFIF IFTIKHAR

Environmental Science Graduate | Graduate Research Applicant

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## RESEARCH INTERESTS

• River Morphodynamics | Channel Migration | Geomorphic Sensitivity | Remote Sensing & Spatial Modeling

## EDUCATION

**Master of Science in Environmental Science** — Expected October 2026
Khulna University, Khulna, Bangladesh

**Bachelor of Science in Environmental Science** — 2023
Khulna University, Khulna, Bangladesh
CGPA: 3.64 / 4.00

## SELECTED RESEARCH PROJECTS

**Research Project | Manuscript in Preparation - Spatially Offset Curvature-Migration Coupling in the Tidal Rupsha-Pasur River Corridor from Multi-Decadal Satellite Observations**

- Reconstructed channel geometry across 35 analysis years (1988-2025) from Landsat imagery using adaptive MNDWI thresholding and topology-aware processing.
- Applied hierarchical Bayesian modeling to 25,923 station-period observations and evaluated measurement uncertainty using empirical error analysis and a 2,000-realization geometry ensemble.

**MSc Dissertation (in progress) - Analyzing Land-Use Transitions and Socio-Economic Conditions in Relation to Morphological Change of the Rupsha and Pasur Rivers in Bangladesh**

- Integrated 1994-2024 Landsat-derived river morphology and LULC transitions with a 200-household assessment of livelihood vulnerability and adaptation.

**BSc Dissertation - Ecosystem Service Accounting of Mongla Upazilla of Bagerhat District in Bangladesh**

- Mapped LULC change in Mongla using Landsat and linked land-use classes to ecosystem-service valuation and government production/revenue data.

## RESEARCH EXPERIENCE

**Research Assistant, Khulna University** - supervised by Sadhon Chandra Swarnokar — Jan 2024-Present

**Geospatial Research - Climate Change Risk, Impact and Vulnerability Assessment** (funded by World Vision)

- Managed remote-sensing, GIS, and georeferenced household datasets for hazard, exposure, and vulnerability analysis; contributed spatial QA and risk-mapping outputs.

**Urban Displacement Research - Satkhira and Bagerhat** (IFRC / Bangladesh Red Crescent Society / American Red Cross)

- Co-developed FGD/KII instruments, conducted thematic analysis, and contributed to research synthesis on climate-linked urban displacement.

**Researcher, Environment and Sustainability Research Initiative (ESRI)** — Feb 2024-Present

- Contributed to questionnaire development, data management, statistical analysis, manuscript drafting, and research documentation across collaborative environmental studies.

## PEER-REVIEWED JOURNAL ARTICLES

1. Parvez, M. S., Roy, S., Ahmed, M. R., **Iftikhar, A.**, Rumi, M. H. M. H., Swarnokar, S. C., & Khan, R. (2026). Spatial distribution, ecological risk, and source apportionment of sedimentary metals in the Sundarbans mangrove ecosystem of Bangladesh. *Marine Pollution Bulletin, 232*, 120042. https://doi.org/10.1016/j.marpolbul.2026.120042

2. Swarnokar, S. C., Mou, S. I., Sharmi, S. D., **Iftikhar, A.**, & Jesmin, S. (2025). Climate-induced risks, adaptation, and mitigation responses: A comparative study on climate-stressed coastal communities. *Frontiers in Climate, 7*, 1553579. https://doi.org/10.3389/fclim.2025.1553579

3. Hasan, S. A., Patwary, M. M., **Iftikhar, A.**, Rahman, M. A., Badhon, M. I. A., Tuna, S. T. B., Browning, M. H. E. M., & Bell, M. L. (2026). Residential greenness and perinatal depression in low- and middle-income countries: A multi-country population-based study. *Environmental Research*, 125647. https://doi.org/10.1016/j.envres.2026.125647

4. Tamim, A. R., Patwary, M. M., Bardhan, M., Badhon, M. I., Rahman, M. S., Sakib, I. C., **Iftikhar, A.**, Kabir, M. P., Pitol, M. N. S., Saha, C., & Browning, M. H. E. M. (2026). Occupational heat risk perceptions and behavioral adaptation strategies among construction and welding workers in Bangladesh. *Scientific Reports, 16*, 22626. https://doi.org/10.1038/s41598-026-53118-x

## CONFERENCE

Ritu, M. R., Patwary, M. M., **Iftikhar, A.**, Badhon, M. I. A., Bardhan, M., Sakib, I. C., Sikder, D., & Kabir, M. P. (2025). Exploring the indoor air pollution reduction behavior among urban residents of Bangladesh using an extended Theory of Planned Behavior (TPB) model. *AGU Fall Meeting Abstracts* (Abstract SY23D-VR8856).

## TECHNICAL AND METHODOLOGICAL SKILLS

**Remote Sensing & GIS:** Multi-temporal Landsat analysis; LULC classification/change detection; MNDWI and adaptive thresholding; channel-geometry reconstruction; riverbank migration; erosion-accretion; spatial overlays; accuracy assessment; ArcGIS 10.8.

**Vulnerability & Quantitative Analysis:** Hierarchical Bayesian modeling; Monte Carlo uncertainty analysis; multiple linear and binary logistic regression; VIF; chi-square/Cramer's V; LVI/LVI-IPCC; sensitivity analysis.

**Research Methods:** Household surveys; sampling; FGD/KII; thematic analysis; georeferenced field data.

**Programming:** Python (applied statistical/Bayesian analysis); PyMC; Google Earth Engine; SPSS; Microsoft Excel.

## PROFESSIONAL EXPERIENCE

**Management Information Systems and Documentation Officer** Oct 2025-Aug 2026
Satkhira Unnayan Sangstha - SMART Project (World Bank / PKSF), Tala, Satkhira

- Managed project databases, M&E; datasets, validation procedures, and analytical reporting.

**Environmental Specialist** Jan-Oct 2025
Geo Smart Planning, Khulna

- Conducted environmental assessments, field data collection, and technical reporting for infrastructure projects.

## SCHOLARSHIPS, AWARDS AND HONOURS

- **National Science and Technology (NST) Fellowship, 2025** - Ministry of Science and Technology, Government of Bangladesh. Competitive national research fellowship supporting postgraduate scientific research.

## CERTIFICATIONS AND PROFESSIONAL TRAINING

- DHS Data Analysis Course with SPSS, R & Google Earth Engine - ESRI, 2025
- Environmental and Social Impact Assessment (ESIA) - BSITBD, 2024

## LEADERSHIP AND EXTRACURRICULAR ACTIVITIES

- Cadet, Bangladesh National Cadet Corps (BNCC), 2020-2023: Paramilitary training in discipline, teamwork, and field coordination.
- Participant, LEAD Bangladesh Programme (British Council), 2021-2022: Leadership training focused on community development initiatives.
- Volunteer, International Conferences on Environment, 2022 & 2024: Supported conference organization and academic networking.

**English Proficiency:** IELTS Academic - Overall Band 7.0 (31 October 2025)